DVS Industry
Test & Validation Report

Field	Value
Document ID	DVS-QA-TR-001
Revision	1.0
Date of execution	24 October 2026
System under test	DVS Industry ERP + Perception Service, branch `master` — baseline `b799e8e`, verified after fixes at `792e48c`
Repository	https://github.com/SANCHI3/DVS_INDUSTRY
Test standard followed	Structure after IEEE 829 / ISO/IEC/IEEE 29119-3 (Test Summary Report)
Overall verdict	**PASS** — 397 / 397 automated tests pass; 4 defects found and closed

1. Purpose and scope

This report records a full regression run of the DVS Industry system and the validation status of its AI models. It covers:

#	Subsystem	Type of test
1	Backend API (Node.js / Express / Prisma)	Static analysis, code-style, unit and integration tests
2	Frontend (React / Vite)	Static analysis, unit tests, production build
3	End-to-end (browser → API → PostgreSQL)	Playwright UI tests against a seeded database
4	Perception service (Python / PyTorch)	Unit, regression and false-positive scenario tests
5	AI models	Held-out evaluation metrics (precision, recall, F1, baseline comparison)
6	Supply chain	`npm audit` of production dependencies

Out of scope: long-duration field trials on the deployment camera (see §8).

2. Test environments

Environment	Where	Configuration
CI — GitHub Actions	`ubuntu-latest` hosted runners	Node 22, Python 3.12, CPU-only PyTorch, PostgreSQL 16 service container
Local workstation	Windows 11, developer laptop	Node 24.14, Python 3.11 venv, CPU-only PyTorch 2.14.1

3. Execution records (where the tests were run)

3.1 GitHub Actions — continuous integration

Workflow: .github/workflows/ci.yml — runs on every push to `master` and on every pull request.

Final verification run — commit `792e48c` (24 Oct 2026, after all fixes): Run #40078158730 — ✅ SUCCESS (5 / 5 jobs)

Job	Result	Duration	Link
backend (typecheck, lint, format, 212 tests)	✅ Pass	52 s	job log
frontend (typecheck, lint, 55 tests, build)	✅ Pass	29 s	job log
e2e (Playwright + PostgreSQL 16)	✅ Pass	1 min 40 s	job log
perception-service (pytest, 130 tests)	✅ Pass	1 min 25 s	job log
audit (npm, production deps)	✅ Pass	7 s	job log

Run history during this test cycle:

Run	Commit	Result	Note
39976459749	`b799e8e` (baseline, 23 Oct)	✅ Success	Before this test cycle
40077004703	`2908ff7`	❌ audit failed	DEF-001…003 fixed; new advisories found (DEF-004)
40077665767	`00b3ad9`	❌ backend, e2e failed	Lockfile written by npm 11, rejected by npm 10 (DEF-004)
40078158730	`792e48c`	✅ Success	All defects closed

Earlier successful runs: 39963165846, 39949564310, 39825397831, 39822663506. Full history: https://github.com/SANCHI3/DVS_INDUSTRY/actions

3.2 Local regression run (24 Oct 2026)

Suite	Command	Result
Backend — typecheck	`cd backend && npm run typecheck`	✅ 0 errors
Backend — lint	`npm run lint`	✅ 0 problems
Backend — code style	`npm run format:check`	❌ → ✅ 4 files failed, fixed (DEF-001)
Backend — tests	`npm test`	✅ **212 / 212 passed** (23 files)
Frontend — typecheck	`cd frontend && npm run typecheck`	✅ 0 errors
Frontend — lint	`npm run lint`	✅ 0 problems
Frontend — tests	`npm test`	✅ **55 / 55 passed** (9 files)
Frontend — production build	`npm run build`	✅ Built in 14.2 s
Perception — tests	`cd perception-service && python -m pytest -q`	❌ → ✅ 1 failed (DEF-002), fixed → **130 / 130 passed**

Suite	Command	Result
Backend — dependency audit	`npm audit --omit=dev --audit-level=high`	❌ → ✅ 3 high / 2 moderate (DEF-004), fixed → 0 vulnerabilities
Model metrics consistency	`python -m evaluation.run_all`	❌ → ✅ exited with error (DEF-003), fixed → 21 result rows regenerated, all stored metrics consistent with confusion counts

4. Summary of results

Suite	Tests	Passed	Failed	Pass rate
Backend (Vitest + Supertest)	212	212	0	100 %
Frontend (Vitest)	55	55	0	100 %
Perception (pytest)	130	130	0	100 %
End-to-end (Playwright, CI)	2 spec files	✅ all	0	100 %
Total (unit + integration)	**397**	**397**	**0**	**100 %**

5. AI model performance (held-out test data)

Source: docs/RESULTS.md, generated by `perception-service/evaluation/run_all.py` from the metrics files saved at training time. The script recomputes precision and recall from the stored confusion counts and fails if they disagree.

Incident	Deployed model	Precision	Recall	F1	Baseline F1	Test data	Status
Fight	R3D-18 3D CNN (video clips)	0.94	0.97	**0.96**	0.85	344 test videos	✅ Accepted
Fall	YOLOv8-Pose + BiGRU	0.86	0.94	**0.90**	0.81	160 videos, 5-fold CV	✅ Accepted
Fire	YOLOv8n + learned verifier	1.00	0.91	**0.95**	0.87	27 FIRESENSE videos	✅ Accepted
Smoke	YOLOv8n + learned verifier	1.00	0.85	**0.92**	0.92	22 FIRESENSE videos	✅ Accepted
Unusual activity	Autoencoder (unsupervised)	0.96	0.67	**0.79**	0.52	102 videos	⚠ Accepted with limitation (recall)

Baseline = the simpler method each model replaced: fight v1 model, box-shape fall rule, YOLO alone, autoencoder v1. The fire and smoke test sets are small (22–27 videos), so their 95 % confidence intervals are wide (see `RESULTS.md` §1).

6. Defect log

ID	Severity	Component	Description	Root cause	Resolution	Status
DEF-001	Low	backend (tests)	`npm run format:check` failed on 4 test files: `assistant.service.test.ts`, `tools.test.ts`, `assistant.routes.test.ts`, `security.kpis.test.ts`	CI did not run the Prettier check, so unformatted code was merged without failing the build	Files reformatted with Prettier; `npm run format:check` added to the backend CI job so it cannot happen again	✅ Closed — re-test passed (212/212, style clean)
DEF-002	Medium	perception-service (evaluation)	`evaluation/ablation.py` line 558 raised `SyntaxError: f-string: unmatched '['` on Python 3.11; `test_evaluation.py::test_ablation_overrides_and_summary` failed	Same quote character reused inside an f-string (`{cal["clips"]}`), which is legal only on Python ≥ 3.12; the README supports 3.10–3.12, but CI tests only 3.12	Inner quotes changed to `{cal['clips']}`; all perception modules compile on 3.11	✅ Closed — re-test passed (130/130)
DEF-003	Low	perception-service (evaluation)	`python -m evaluation.run_all` exited with `metrics files neither used nor excluded: ['scene_anomaly_ae_v3_factory.metrics.json']`, so `RESULTS.md` could not be regenerated	The v3 factory-footage model (added 2026-10) is reported in `RESULTS.md` §7 by its own script but was not registered with the collector	File registered in the collector's exclusion list with its reason; outputs regenerated, no metric changed	✅ Closed — re-run passed (21 rows)
DEF-004	High	backend (dependencies)	CI `audit` job failed on 24 Oct 2026 (run 40077004703): new advisories for production dependencies — `multer` 2.4.0 (DoS, GHSA-q7qp-rjg7-q26m), `zod` < 4.6.6 (memory exhaustion, GHSA-mrhx-c7xc-m3w5), `nodemailer` ≤ 10.1.3 (address parsing, 3 advisories), `body-parser` 2.3.0, `ip-address` ≤ 10.7.3	Advisories disclosed after the previous green run; no code change on our side. A first fix made with npm 11 rewrote the lockfile in a form CI's npm 10 rejects (`npm ci`: "Missing: magicast@0.3.5"), failing the backend and e2e jobs (run 40077665767)	Lockfile restored and the five packages updated with npm 10 (as in CI), within existing semver ranges: multer 2.5.0, zod 4.6.7, nodemailer 10.2.1, body-parser 2.4.0, ip-address 10.7.4 (42-line lockfile change)	✅ Closed — 0 production vulnerabilities, `npm ci` in sync, 212/212 tests pass

No defect affected the live system or the reported model metrics.

7. Test coverage by area (perception service)

Area	Test file(s)
Rule engine and confirmation tiers	`test_rules_logic.py`, `test_crowd_logic.py`
False-positive regression (18 scenarios)	`test_false_positives.py`
Fight model sampling and gating	`test_rgb_sampling.py`, `test_learned_models.py`
Fall model	`test_fall_model.py`
Fire / smoke verifier	`test_fire_verifier.py`
Unusual-activity autoencoder	`test_scene_anomaly.py`, `test_train_factory.py`
Evidence extraction	`test_evidence.py`
Incident API, schema, de-duplication, outbox	`test_incident_api.py`, `test_incident_dedup.py`, `test_incident_report.py`, `test_reporter_outbox.py`, `test_dashboard_client.py`
Video input	`test_cctv_input.py`, `test_stream_reader_file.py`, `test_demo_mode.py`
LLM / plain-language output	`test_plain_language.py`
Evaluation statistics (bootstrap, McNemar)	`test_evaluation.py`

8. Known limitations and residual risk

#	Limitation	Impact	Mitigation
L1	False alarms per hour on the deployment camera are not yet measured; on proxy clips the deployed system gives 37.6 false alarms/h	Operator alert fatigue possible in the field	`python -m evaluation.false_alarms --video <recording>` is ready to run once field footage exists
L2	Unusual-activity recall is 0.67	About 1 in 3 abnormal events not flagged by this detector	Specific detectors (fight, fall, fire) cover the high-severity events
L3	Fire/smoke test sets are small (27 / 22 videos)	Wide confidence intervals	Collect more site footage
L4	Fight interaction check separates classes weakly (ROC AUC 0.55)	Used as a review signal, not a veto	Re-tune on field recordings

9. Conclusion

All automated test suites pass locally and in GitHub Actions. Four defects were found and fixed (backend code style; a Python 3.11 syntax error in the ablation report; a results-collector error; newly disclosed vulnerabilities in five backend dependencies). Lockfiles must be regenerated with the npm version CI uses (npm 10 / Node 22); a CI step was added so the style defect cannot recur. Recommendation: add Python 3.10/3.11 to the CI matrix, since the README claims support for them. The deployed AI models meet or beat their baselines on held-out data, with the limitations listed in §8. The build is **released for demonstration**.

10. How to reproduce

```
# Backend
cd backend && npm ci && npx prisma generate
npm run typecheck && npm run lint && npm run format:check && npm test

# Frontend
cd ../frontend && npm ci
npm run typecheck && npm run lint && npm test && npm run build

# Perception service
cd ../perception-service
pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
pip install -r requirements.txt
python -m pytest -q
python -m evaluation.run_all
```

Approval

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